

# GATE DA (Data Science & AI) Practice Question Bank - Set 3

7 Topics · Previous-Year-Style + New Practice Questions · Full Solutions

This set covers seven topics: Continuous Random Variables & PDF/CDF, Regularization (Ridge/Lasso), Relational Algebra Operations, Profit, Loss & Discount, Introduction to Complexity (Big-O/Theta/Omega), Solving Recurrences via the Substitution Method, and Uninformed Search -- DFS (mechanism & properties).

SAME HONESTY POLICY AS BEFORE: GATE DA has run only in 2024, 2025 and 2026, and it is a non-coding, MCQ/MSQ/NAT written exam. Every "old" question below is tagged with its real provenance:

[VERIFIED PYQ] -- a genuine, well-documented past GATE question (year and paper stated). Where the source is GATE CSE rather than GATE DA specifically, that is stated explicitly -- these test the identical concept/format used in GATE DA's shared syllabus.

[GATE-style] -- not a verbatim past question (either no verified PYQ exists on that exact sub-point, or the original figure/data could not be reliably reproduced), but written to match real GATE DA difficulty, phrasing, and mark-weightage.

Section B ("New Practice Questions") in every topic is freshly written GATE-level practice, as requested -- none of these are claimed to be past questions.

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# Topic 1: Continuous Random Variable & PDF/CDF

## Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE DA 2025]

### Q1. [NAT · 2 marks]

Let  $X$  be a continuous random variable whose CDF  $F_X(x)$ , for some  $t$ , is given by  $F_X(x) = 0$  for  $x \leq t$ ;  $(x-t)/(4-t)$  for  $t \leq x \leq 4$ ; and 1 for  $x \geq 4$ . If the median of  $X$  is 3, find the value of  $t$ .

**Answer: 2**

**Solution:** At the median,  $F_X(3) = 0.5$ . So  $(3-t)/(4-t) = 0.5 \Rightarrow 3-t = 0.5(4-t) = 2-0.5t \Rightarrow 3-t-2+0.5t = 0 \Rightarrow 1-0.5t = 0 \Rightarrow t = 2$ . (Check:  $(3-2)/(4-2) = 1/2$  ✓)

[GATE-style]

### Q2. [NAT · 1 mark]

A continuous random variable  $X$  has PDF  $f(x) = kx^2$  for  $0 < x < 1$ , and 0 elsewhere. Find  $k$ .

**Answer: 3**

**Solution:**  $\int_0^1 kx^2 dx = k/3 = 1 \Rightarrow k = 3$ .

[GATE-style]

### Q3. [NAT · 2 marks]

Using  $k=3$  from the previous question ( $f(x)=3x^2$  for  $0 < x < 1$ ), find  $P(X \leq 0.5)$ .

**Answer: 0.125**

**Solution:**  $P(X \leq 0.5) = \int_0^{0.5} 3x^2 dx = [x^3]_0^{0.5} = 0.125$ .

[GATE-style]

### Q4. [NAT · 1 mark]

For  $X \sim \text{Uniform}(2,8)$ , find the constant value of the PDF  $f(x)$  over its support (correct to 4 decimals).

**Answer: 0.1667**

**Solution:**  $f(x) = 1/(b-a) = 1/(8-2) = 1/6 \approx 0.1667$ .

[GATE-style]

### Q5. [NAT · 1 mark]

For  $X \sim \text{Uniform}(2,8)$ , find  $P(3 < X < 5)$  (correct to 2 decimals).

**Answer: 0.33**

**Solution:** For a uniform distribution,  $P(3 < X < 5) = (5-3)/(8-2) = 2/6 \approx 0.33$ .

[GATE-style]

### Q6. [MCQ · 1 mark]

Which of the following is TRUE for the PDF  $f(x)$  of any continuous random variable  $X$ ?

- (A)  $f(x)$  is always  $\leq 1$  for all  $x$
- (B)  $f(x)$  directly represents  $P(X=x)$
- (C)  $\int f(x)dx$  over the entire real line equals 1, and  $f(x) \geq 0$  everywhere
- (D)  $f(x)$  must be a strictly decreasing function

**Answer: (C)**

[GATE-style]

**Q7. [MSQ · 2 marks]**

Which of the following are TRUE about the CDF  $F(x)$  of a continuous random variable?

- (A)  $F(x)$  is non-decreasing
- (B)  $F(x)$  is continuous (no jumps), unlike the discrete case
- (C)  $F'(x) = f(x)$  wherever  $f$  is continuous
- (D)  $F(-\infty) = 1$

**Answer: (A), (B), (C)**

**Solution:** (D) is false --  $F(x)$  must approach 0, not 1, as  $x \rightarrow -\infty$ .

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[GATE-style]

**Q8. [NAT · 2 marks]**

$X$  follows an exponential distribution with rate  $\lambda=2$ , i.e.  $f(x)=2e^{-2x}$  for  $x \geq 0$ . Find  $P(X > 1)$  (correct to 4 decimals).

**Answer: 0.1353**

**Solution:** For exponential( $\lambda$ ),  $P(X>a) = e^{-\lambda a}$ . So  $P(X>1) = e^{-2} \approx 0.1353$ .

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[GATE-style]

**Q9. [NAT · 1 mark]**

A continuous random variable has PDF  $f(x) = 1/2$  for  $0 \leq x \leq 2$  (uniform), and 0 elsewhere. Find  $E[X]$ .

**Answer: 1**

**Solution:** For Uniform( $a,b$ ),  $E[X] = (a+b)/2 = (0+2)/2 = 1$ .

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[GATE-style]

**Q10. [MCQ · 1 mark]**

For any continuous random variable  $X$  and any specific real value  $c$ ,  $P(X=c)$  equals:

- (A)  $f(c)$
- (B) 0
- (C)  $F(c)$
- (D)  $1-F(c)$

**Answer: (B)**

**Solution:** For continuous random variables, the probability of any single exact point is always 0 -- probability is only meaningfully assigned to intervals.

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## Section B — New Practice Questions (10)

[New Practice]

**Q11. [NAT · 2 marks]**

A continuous RV  $X$  has PDF  $f(x) = 2x$  for  $0 \leq x \leq 1$ . Find  $P(X > 0.5)$ .

**Answer: 0.75**

**Solution:**  $P(X>0.5) = \int_{0.5}^1 2x \, dx = [x^2]_{0.5}^1 = 1 - 0.25 = 0.75$ .

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[New Practice]

**Q12. [NAT · 2 marks]**

Using  $f(x)=2x$  for  $0 \leq x \leq 1$  from the previous question, find  $E[X]$  (correct to 2 decimals).

**Answer: 0.67**

**Solution:**  $E[X] = \int_0^1 x \cdot 2x \, dx = \int_0^1 2x^2 \, dx = (2/3)x^3 \Big|_0^1 = 2/3 \approx 0.67.$

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[New Practice]

**Q13. [MCQ · 1 mark]**

A continuous RV has CDF  $F(x) = x^2$  for  $0 \leq x \leq 1$  (0 for  $x < 0$ , 1 for  $x > 1$ ). What is the PDF  $f(x)$  for  $0 < x < 1$ ?

- (A)  $2x$
- (B)  $x^2$
- (C) 1
- (D)  $x$

**Answer: (A)**

**Solution:**  $f(x) = F'(x) = d/dx(x^2) = 2x.$

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[New Practice]

**Q14. [NAT · 2 marks]**

Using  $F(x) = x^2$  ( $0 \leq x \leq 1$ ) from the previous question, find  $P(0.2 < X < 0.6)$ .

**Answer: 0.32**

**Solution:**  $P(0.2 < X < 0.6) = F(0.6) - F(0.2) = 0.36 - 0.04 = 0.32.$

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[New Practice]

**Q15. [MCQ · 1 mark]**

For  $X \sim \text{Uniform}(a, b)$ , the variance is:

- (A)  $(b-a)/2$
- (B)  $(b-a)^2/12$
- (C)  $(b-a)^2/4$
- (D)  $(b+a)/2$

**Answer: (B)**

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[New Practice]

**Q16. [NAT · 1 mark]**

Using the formula from the previous question, find  $\text{Var}(X)$  for  $X \sim \text{Uniform}(0, 10)$ .

**Answer: 8.33**

**Solution:**  $\text{Var}(X) = (10-0)^2/12 = 100/12 \approx 8.33.$

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[New Practice]

**Q17. [MSQ · 2 marks]**

Which of the following are properties a function  $f(x)$  must satisfy to be a valid PDF over its support?

- (A)  $f(x) \geq 0$  everywhere
- (B) The total area under  $f(x)$  equals 1
- (C)  $f(x)$  must be continuous everywhere (no jump discontinuities allowed)
- (D)  $f(x)$  can exceed 1 at some points

**Answer: (A), (B), (D)**

**Solution:** (C) is false -- a PDF is allowed to have jump discontinuities (e.g. a piecewise-defined uniform density) and still be perfectly valid.

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[New Practice]

**Q18. [NAT · 1 mark]**

X follows an exponential distribution with rate  $\lambda = 0.5$ . Find  $E[X]$ .

**Answer: 2**

**Solution:** For exponential( $\lambda$ ),  $E[X] = 1/\lambda = 1/0.5 = 2$ .

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*[New Practice]*

**Q19. [MCQ · 1 mark]**

If  $F(x)$  is the CDF of a continuous random variable  $X$ , then  $P(a < X \leq b)$  equals:

- (A)  $F(b) - F(a)$
- (B)  $F(a) - F(b)$
- (C)  $F(a) + F(b)$
- (D)  $F(b) \times F(a)$

**Answer: (A)**

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*[New Practice]*

**Q20. [NAT · 2 marks]**

X has PDF  $f(x) = 3x^2$  for  $0 \leq x \leq 1$ . Find the median  $m$ , i.e. the value such that  $F(m)=0.5$  (correct to 2 decimals).

**Answer: 0.79**

**Solution:**  $F(x) = x^3$ . Setting  $x^3 = 0.5$  gives  $x = 0.5^{1/3} \approx 0.7937$  (accept 0.79–0.80).

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## Topic 2: Regularization – Ridge & Lasso

### Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE DA 2024, Official Sample Question Paper, Question 12]

#### Q1. [MCQ · 2 marks]

Increasing the regularization coefficient value for a Ridge regressor will: (i) Increase or maintain model bias. (ii) Decrease model bias. (iii) Increase or maintain model variance. (iv) Decrease model variance. Which combination is correct?

- (A) i and iii
- (B) i and iv
- (C) ii and iii
- (D) ii and iv

**Answer: (B)**

**Solution:** A larger regularization coefficient forces the model toward simpler solutions (smaller weight magnitudes). This restricts the model's flexibility to fit the training data exactly, which increases (or at least maintains) bias (i), while making the model less sensitive to the particular training sample, which decreases variance (iv).

[GATE-style]

#### Q2. [MCQ · 1 mark]

The Ridge regression cost function is:

- (A)  $\sum (y_{xii} - \hat{y}_{xii})^2 + \lambda \sum |w_j|$
- (B)  $\sum (y_{xii} - \hat{y}_{xii})^2 + \lambda \sum w_j^2$
- (C)  $\sum |y_{xii} - \hat{y}_{xii}| + \lambda \sum w_j^2$
- (D)  $\sum (y_{xii} - \hat{y}_{xii})^2$  only, with no penalty term

**Answer: (B)**

[GATE-style]

#### Q3. [MCQ · 1 mark]

The Lasso regression cost function is:

- (A)  $\sum (y_{xii} - \hat{y}_{xii})^2 + \lambda \sum |w_j|$
- (B)  $\sum (y_{xii} - \hat{y}_{xii})^2 + \lambda \sum w_j^2$
- (C)  $\sum (y_{xii} - \hat{y}_{xii})^2$  only
- (D)  $\sum (y_{xii} - \hat{y}_{xii})^2 / \lambda$

**Answer: (A)**

[GATE-style]

#### Q4. [MCQ · 1 mark]

Which regularization technique is more likely to produce a sparse model (some weights exactly zero)?

- (A) Ridge (L2)
- (B) Lasso (L1)
- (C) Both equally
- (D) Neither

**Answer: (B)**

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[GATE-style]

**Q5. [MSQ · 2 marks]**

Which of the following are TRUE about Ridge regression (L2 regularization)?

- (A) It shrinks coefficients toward zero but typically does not set them exactly to zero
- (B) It adds a penalty proportional to the sum of squared coefficients
- (C) As  $\lambda \rightarrow 0$ , ridge regression reduces to ordinary least squares
- (D) It performs automatic feature selection by driving irrelevant coefficients exactly to zero

**Answer: (A), (B), (C)**

**Solution:** (D) is false -- exact zeroing-out of coefficients (automatic feature selection) is the characteristic property of Lasso, not Ridge.

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[GATE-style]

**Q6. [NAT · 1 mark]**

For Ridge regression with cost  $J(w) = \sum (y_{xii} - wx_{xii})^2 + \lambda w^2$ , if  $\lambda$  is increased without bound ( $\lambda \rightarrow \infty$ ), what value does the optimal  $w$  tend toward?

**Answer: 0**

**Solution:** As  $\lambda \rightarrow \infty$ , the penalty term dominates the cost function, forcing the optimal  $w$  toward 0 (the model becomes maximally simple).

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[GATE-style]

**Q7. [MCQ · 1 mark]**

In two dimensions (weights  $w_1, w_2$ ), the geometric shape of the Lasso constraint region ( $|w_1| + |w_2| \leq t$ ) is a:

- (A) Circle
- (B) Diamond (a rotated square)
- (C) Ellipse
- (D) Straight line

**Answer: (B)**

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[GATE-style]

**Q8. [MCQ · 1 mark]**

In two dimensions, the geometric shape of the Ridge constraint region ( $w_1^2 + w_2^2 \leq t$ ) is a:

- (A) Diamond
- (B) Circle
- (C) Triangle
- (D) Straight line

**Answer: (B)**

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[GATE-style]

**Q9. [MCQ · 1 mark]**

If the regularization strength  $\lambda$  is set extremely high in either Ridge or Lasso, the resulting model tends toward:

- (A) Perfect overfitting
- (B) A trivial, overly simple model with most/all coefficients driven toward zero (underfitting)
- (C) The exact OLS solution

(D) An undefined result

**Answer: (B)**

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[GATE-style]

**Q10. [MSQ · 2 marks]**

Which of the following are TRUE about the effect of regularization on the bias-variance tradeoff?

- (A) Increasing  $\lambda$  generally increases bias
- (B) Increasing  $\lambda$  generally decreases variance
- (C) Regularization eliminates all bias while still reducing variance
- (D) There is generally an optimal  $\lambda$  that minimizes total expected test error ( $\text{bias}^2 + \text{variance}$ )

**Answer: (A), (B), (D)**

**Solution:** (C) is false -- regularization deliberately introduces (or increases) bias in exchange for reduced variance; it does not eliminate bias.

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## Section B — New Practice Questions (10)

[New Practice]

**Q11. [MCQ · 1 mark]**

Elastic Net regularization combines:

- (A) Only the L1 (Lasso) penalty
- (B) Only the L2 (Ridge) penalty
- (C) Both L1 and L2 penalties together
- (D) Neither penalty

**Answer: (C)**

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[New Practice]

**Q12. [NAT · 2 marks]**

For single-feature Ridge regression,  $w = \sum x_i y_i / (\sum x_i^2 + \lambda)$ . Given  $\sum x_i y_i = 20$ ,  $\sum x_i^2 = 8$ , and  $\lambda = 2$ , find  $w$ .

**Answer: 2**

**Solution:**  $w = 20/(8+2) = 20/10 = 2$ .

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[New Practice]

**Q13. [MCQ · 1 mark]**

Compared to OLS, Ridge regression coefficients are typically:

- (A) Larger in magnitude
- (B) Smaller in magnitude (shrunk toward zero)
- (C) Identical to OLS
- (D) Always negative

**Answer: (B)**

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[New Practice]

**Q14. [MSQ · 2 marks]**

Which of the following are TRUE about Lasso regression?

- (A) It can perform feature selection by setting some coefficients exactly to zero

- (B) Its L1 penalty term is differentiable everywhere, including at  $w=0$
- (C) It uses a penalty of the form  $\lambda \sum |w_j|$
- (D) As  $\lambda \rightarrow 0$ , Lasso reduces to ordinary least squares

**Answer: (A), (C), (D)**

**Solution:** (B) is false -- the L1 penalty  $|w|$  has a non-differentiable "kink" exactly at  $w=0$ , and this kink is precisely what allows Lasso to drive coefficients to exact zero.

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*[New Practice]*

**Q15. [MCQ · 1 mark]**

Which regularization method is generally preferred when you believe most features are irrelevant and want automatic feature selection?

- (A) Ridge
- (B) Lasso
- (C) No regularization
- (D) Increasing the learning rate

**Answer: (B)**

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*[New Practice]*

**Q16. [MCQ · 1 mark]**

Which regularization method is generally preferred when features are highly correlated (multicollinearity), and you want to retain all of them but shrink their influence?

- (A) Ridge
- (B) Lasso
- (C) No regularization
- (D) Increasing the learning rate

**Answer: (A)**

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*[New Practice]*

**Q17. [MCQ · 1 mark]**

If  $\lambda$  in a Lasso model is doubled (with the data unchanged), the number of exactly-zero coefficients typically:

- (A) Increases (or stays the same)
- (B) Always decreases
- (C) Is always exactly halved
- (D) Is unrelated to  $\lambda$

**Answer: (A)**

**Solution:** A larger  $\lambda$  imposes a stronger penalty on non-zero coefficients, typically driving more of them to exactly zero.

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*[New Practice]*

**Q18. [MCQ · 1 mark]**

Regularization is primarily added to a model's cost function in order to:

- (A) Increase training accuracy at all costs
- (B) Reduce overfitting by penalizing model complexity (large weights)
- (C) Directly speed up gradient descent's convergence rate
- (D) Increase the number of features used

**Answer: (B)**

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*[New Practice]*

**Q19. [MCQ · 1 mark]**

In Ridge regression, the intercept term (bias,  $w_0$ ) is conventionally:

- (A) Regularized (penalized) exactly like all other weights
- (B) Excluded from the regularization penalty
- (C) Always set to zero
- (D) The only term that is regularized

**Answer: (B)**

**Solution:** Standard practice excludes the intercept from the penalty, since shrinking the baseline prediction level is not the goal of regularization.

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*[New Practice]*

**Q20. [NAT · 2 marks]**

For Ridge regression,  $w = \sum x_i y_i / (\sum x_i^2 + \lambda)$ , with  $\sum x_i y_i = 15$ ,  $\sum x_i^2 = 5$ ,  $\lambda = 5$ , find  $w$ .

**Answer: 1.5**

**Solution:**  $w = 15/(5+5) = 15/10 = 1.5$ .

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## Topic 3: Relational Algebra - Operations

### Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CSE 2024, Set 2]

#### Q1. [MCQ · 2 marks]

The relation schema Person(pid, city) describes the city of residence for every person, uniquely identified by pid. Using only selection, projection, cross product, and rename, what is the minimum number of cross-product operations needed to find the list of cities where at least 3 persons reside?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

#### Answer: (B)

**Solution:** Without aggregate/grouping operators, detecting "at least 3 persons sharing a city" requires comparing three renamed copies of Person pairwise (ensuring their pids are distinct while their cities match). Combining three relations requires exactly 2 cross-product operations: (copy1  $\times$  copy2)  $\times$  copy3, followed by appropriate selections and a projection on city.

[VERIFIED PYQ — GATE CSE 2006, classic natural-join sizing question]

#### Q2. [NAT · 2 marks]

Relations  $r_1(P, Q, R)$  and  $r_2(R, S, T)$  have primary keys P and R respectively.  $r_1$  contains 2000 tuples and  $r_2$  contains 2500 tuples. What is the maximum possible number of tuples in the natural join  $r_1 \bowtie r_2$  (joined on the common attribute R)?

#### Answer: 2000

**Solution:** Since R is the primary key of  $r_2$ , each R-value appears at most once in  $r_2$ . So every tuple of  $r_1$  can match with at most one tuple of  $r_2$ , giving a maximum join size equal to the number of tuples in  $r_1$  (achieved when every R-value of  $r_1$  exists in  $r_2$ ), i.e. 2000.

[VERIFIED PYQ — GATE CSE 1999, classic]

#### Q3. [MCQ · 1 mark]

Consider the join of relation R (with m tuples) and relation S (with n tuples), with no key or referential constraints assumed. What are the maximum and minimum possible sizes of the join  $R \bowtie S$ , respectively?

- (A)  $m \times n$  and 0
- (B)  $m + n$  and 0
- (C)  $m \times n$  and  $\min(m, n)$
- (D)  $\max(m, n)$  and 0

#### Answer: (A)

**Solution:** With no constraints, in the best case for matching, every tuple of R could match every tuple of S (a full Cartesian product), giving up to  $m \times n$  tuples; if no tuples satisfy the join condition at all, the join can have as few as 0 tuples.

[GATE-style]

#### Q4. [MCQ · 1 mark]

The selection operation  $\sigma_{\text{condition}}(R)$  in relational algebra:

- (A) Selects a subset of columns (attributes) of R

- (B) Selects a subset of rows (tuples) of R satisfying the condition
- (C) Combines two relations into one
- (D) Removes duplicate rows from R

**Answer: (B)**

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[GATE-style]

**Q5. [MCQ · 1 mark]**

The projection operation  $\pi_{A1,A2}(R)$ :

- (A) Selects rows of R matching a given condition
- (B) Selects only the specified columns (attributes) A1, A2 of R, removing duplicate resulting rows
- (C) Computes the Cartesian product of R with itself
- (D) Sorts R by attributes A1, A2

**Answer: (B)**

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[GATE-style]

**Q6. [MCQ · 1 mark]**

For the union operation  $R \cup S$  to be valid in relational algebra, R and S must be:

- (A) Union-compatible (same degree, with compatible corresponding domains)
- (B) Identical relations
- (C) Joined on a common key
- (D) Sorted in the same order

**Answer: (A)**

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[GATE-style]

**Q7. [MCQ · 1 mark]**

The natural join  $R \bowtie S$  automatically:

- (A) Computes the Cartesian product with no join condition at all
- (B) Joins R and S on all of their common attribute names, and removes the duplicate copies of those shared columns from the result
- (C) Requires R and S to share no attributes at all
- (D) Behaves identically to the union operation

**Answer: (B)**

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[GATE-style]

**Q8. [NAT · 1 mark]**

Relation R has 5 tuples and relation S has 4 tuples, and R and S share no common attributes. How many tuples are in the Cartesian product  $R \times S$ ?

**Answer: 20**

**Solution:**  $|R \times S| = |R| \times |S| = 5 \times 4 = 20$ .

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[GATE-style]

**Q9. [MSQ · 2 marks]**

Which of the following are considered fundamental (primitive) operations of relational algebra, from which other operations (like natural join) can be derived?

- (A) Selection ( $\sigma$ )

- (B) Projection ( $\pi$ )
- (C) Union ( $\cup$ )
- (D) Natural Join ( $\bowtie$ )

**Answer: (A), (B), (C)**

**Solution:** (D) Natural join -- along with set intersection and division -- is a derived operation, expressible in terms of the primitive operations (selection, projection, union, set difference, Cartesian product, and rename).

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[GATE-style]

**Q10. [MCQ · 1 mark]**

The set difference operation  $R - S$  returns:

- (A) Tuples that are in both R and S
- (B) Tuples that are in R but NOT in S
- (C) Tuples that are in S but NOT in R
- (D) All tuples from both R and S combined

**Answer: (B)**

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## Section B — New Practice Questions (10)

[New Practice]

**Q11. [NAT · 1 mark]**

Relation Employee has 6 attributes. Applying  $\pi_{\text{Name, Salary}}(\text{Employee})$  returns a relation with how many attributes?

**Answer: 2**

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[New Practice]

**Q12. [MCQ · 1 mark]**

If relation R has r tuples and relation S has s tuples, and they are union-compatible, what is the maximum possible number of tuples in  $R \cup S$ ?

- (A)  $r+s$
- (B)  $r \times s$
- (C)  $\max(r, s)$
- (D)  $\min(r, s)$

**Answer: (A)**

**Solution:** The union can have at most  $r+s$  tuples (if there is no overlap at all); duplicates are removed, so the actual result could have fewer, but the maximum is  $r+s$ .

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[New Practice]

**Q13. [MCQ · 1 mark]**

The theta-join  $R \bowtie_{\theta} S$  (a join with a general condition  $\theta$ ) can be expressed using which two primitive operations?

- (A) Selection and Projection
- (B) Selection applied to a Cartesian product:  $\sigma_{\theta}(R \times S)$
- (C) Union and Set Difference
- (D) Projection and Rename

**Answer: (B)**

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[New Practice]

**Q14. [NAT · 2 marks]**

Relation R(A,B) has 3 tuples: (1,2), (1,3), (2,2). Applying  $\pi_A(R)$  (projection on A only, removing duplicates), how many tuples are in the result?

**Answer: 2**

**Solution:** The distinct A values are {1, 2}, so the projection has 2 tuples.

---

[New Practice]

**Q15. [MSQ · 2 marks]**

Which of the following relational algebra operations require the two input relations to be union-compatible (same degree and compatible domains)?

- (A) Union ( $\cup$ )
- (B) Set Difference ( $-$ )
- (C) Intersection ( $\cap$ )
- (D) Cartesian Product ( $\times$ )

**Answer: (A), (B), (C)**

**Solution:** (D) Cartesian product does NOT require union-compatibility -- it can combine relations with entirely different schemas.

---

[New Practice]

**Q16. [MCQ · 1 mark]**

The rename operation  $\rho$  in relational algebra is primarily used to:

- (A) Delete a relation
- (B) Rename a relation and/or its attributes -- useful, e.g., when joining a relation with itself
- (C) Sort the tuples of a relation
- (D) Filter out unwanted rows

**Answer: (B)**

---

[New Practice]

**Q17. [NAT · 1 mark]**

Relation R has 4 tuples and relation S has 4 tuples, and they are identical (same schema, same tuples). How many tuples are in  $R \cap S$ ?

**Answer: 4**

---

[New Practice]

**Q18. [MCQ · 2 marks]**

Which expression correctly computes  $R \cap S$  using only primitive relational algebra operations?

- (A)  $R \cup S$
- (B)  $R - (R - S)$
- (C)  $R \times S$
- (D)  $\sigma_{R=S}(R)$

**Answer: (B)**

**Solution:**  $R \cap S = R - (R - S)$  is the standard derivation of intersection from set difference.

---

[New Practice]

**Q19. [MCQ · 2 marks]**

If relations  $R(A,B)$  and  $S(B,C)$  with  $m$  and  $n$  tuples respectively are combined via natural join  $R \bowtie S$ , and  $B$  is a non-key attribute in both, the size of  $R \bowtie S$  in the worst case can be:

- (A) At most  $m+n$
- (B) At most  $m \times n$
- (C) Exactly  $m \times n$  always
- (D) At most  $\min(m,n)$

**Answer: (B)**

**Solution:** In the worst case (e.g. every  $B$ -value is identical across both relations), every tuple of  $R$  matches every tuple of  $S$ , giving up to  $m \times n$  result tuples.

---

[New Practice]

**Q20. [NAT · 1 mark]**

Relation  $\text{Student}(\text{RollNo}, \text{Name}, \text{Marks})$  has 50 tuples. Applying  $\sigma_{\text{Marks} > 80}(\text{Student})$  returns 12 tuples. How many tuples satisfy  $\text{Marks} \leq 80$ ?

**Answer: 38**

**Solution:**  $50 - 12 = 38$ .

---

## Topic 4: Profit, Loss & Discount

### Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

#### Q1. [NAT · 1 mark]

A shopkeeper buys an item for Rs. 400 and sells it for Rs. 500. Find the profit percentage.

**Answer: 25**

**Solution:** Profit = 100. Profit% =  $(100/400) \times 100 = 25\%$ .

---

[GATE-style]

#### Q2. [NAT · 1 mark]

An item is sold at a loss of 20% for Rs. 640. Find the cost price.

**Answer: 800**

**Solution:**  $CP \times 0.8 = 640 \Rightarrow CP = 640/0.8 = 800$ .

---

[GATE-style]

#### Q3. [MCQ · 2 marks]

A shopkeeper marks an item 40% above cost price and then gives a discount of 10% on the marked price. Find the overall profit percentage.

- (A) 26%
- (B) 30%
- (C) 25%
- (D) 24%

**Answer: (A)**

**Solution:** Let  $CP=100$ .  $MP = 140$ .  $SP = 140 \times 0.9 = 126$ . Profit% =  $126 - 100 = 26\%$ .

---

[GATE-style]

#### Q4. [NAT · 1 mark]

The marked price of an article is Rs. 1200. After a discount of 15%, find the selling price.

**Answer: 1020**

**Solution:**  $SP = 1200 \times 0.85 = 1020$ .

---

[GATE-style]

#### Q5. [MCQ · 2 marks]

If the cost price of 12 articles equals the selling price of 10 articles, find the profit percentage.

- (A) 16.67%
- (B) 20%
- (C) 25%
- (D) 10%

**Answer: (B)**

**Solution:** Let CP of 1 article = 1.  $CP \text{ of } 12 = 12 = SP \text{ of } 10 \Rightarrow SP \text{ of } 1 = 1.2$ . Profit% =  $(1.2 - 1)/1 \times 100 = 20\%$ .

---

[GATE-style]

#### Q6. [NAT · 2 marks]

A trader sells an article at a 10% loss. If he had sold it for Rs. 45 more, he would have made a profit of 5%. Find the cost price.

**Answer: 300**

**Solution:** Let CP = x.  $SP_{\text{loss}} = 0.9x$ .  $SP_{\text{profit}} = 1.05x$ . Difference =  $0.15x = 45 \Rightarrow x = 300$ .

---

[GATE-style]

**Q7. [MCQ · 1 mark]**

Successive discounts of 20% and 10% on a marked price of Rs. 1000 result in a final selling price of:

- (A) Rs. 700
- (B) Rs. 720
- (C) Rs. 730
- (D) Rs. 710

**Answer: (B)**

**Solution:**  $SP = 1000 \times 0.8 \times 0.9 = 720$ .

---

[GATE-style]

**Q8. [MSQ · 2 marks]**

Which of the following statements about profit, loss, and discount are TRUE?

- (A) Discount is always calculated on the marked (listed) price, not the cost price
- (B) Profit percentage is calculated as  $(\text{Profit}/\text{Cost Price}) \times 100$
- (C) Two successive discounts of x% and y% are equivalent to a single discount of exactly  $(x+y)\%$
- (D) Marked price is generally set higher than cost price to allow room for discounts while still yielding profit

**Answer: (A), (B), (D)**

**Solution:** (C) is false -- successive discounts of x% and y% combine to an effective single discount of  $x+y-(xy/100)\%$ , which is less than  $x+y\%$ .

---

[GATE-style]

**Q9. [NAT · 1 mark]**

Find the single equivalent discount for two successive discounts of 20% and 25%.

**Answer: 40**

**Solution:** Equivalent discount =  $20+25-(20 \times 25)/100 = 45-5 = 40\%$ .

---

[GATE-style]

**Q10. [MCQ · 2 marks]**

A shopkeeper wants a 20% profit after giving a 10% discount on the marked price. If the cost price is Rs. 900, what should the marked price be?

- (A) Rs. 1200
- (B) Rs. 1188
- (C) Rs. 1100
- (D) Rs. 1150

**Answer: (A)**

**Solution:** Required SP =  $900 \times 1.2 = 1080$ .  $MP \times 0.9 = 1080 \Rightarrow MP = 1080/0.9 = 1200$ .

---

## Section B — New Practice Questions (10)

[New Practice]

**Q11. [NAT · 1 mark]**

An article is bought for Rs. 750 and sold for Rs. 900. Find the profit percentage.

**Answer: 20**

**Solution:** Profit = 150. Profit% =  $(150/750) \times 100 = 20\%$ .

---

[New Practice]

**Q12. [NAT · 1 mark]**

An article is sold at a loss of 15% for Rs. 1020. Find the cost price.

**Answer: 1200**

**Solution:**  $CP \times 0.85 = 1020 \Rightarrow CP = 1200$ .

---

[New Practice]

**Q13. [MCQ · 1 mark]**

The marked price of an item is Rs. 2000. A discount of 25% is given. Find the selling price.

- (A) Rs. 1500
- (B) Rs. 1400
- (C) Rs. 1600
- (D) Rs. 1450

**Answer: (A)**

**Solution:**  $SP = 2000 \times 0.75 = 1500$ .

---

[New Practice]

**Q14. [NAT · 2 marks]**

The cost price of 15 pens equals the selling price of 12 pens. Find the profit percentage.

**Answer: 25**

**Solution:** Let CP per pen = 1. CP of 15 = 15 = SP of 12  $\Rightarrow$  SP per pen =  $15/12 = 1.25$ . Profit% =  $(1.25 - 1)/1 \times 100 = 25\%$ .

---

[New Practice]

**Q15. [MSQ · 2 marks]**

Which of the following are TRUE regarding profit and loss calculations?

- (A) Loss percentage is calculated as  $(\text{Loss}/\text{Cost Price}) \times 100$
- (B) If Selling Price > Cost Price, there is a profit
- (C) If Selling Price < Cost Price, there is a profit
- (D) Profit or loss percentage is always calculated with Cost Price as the base, not Selling Price

**Answer: (A), (B), (D)**

---

[New Practice]

**Q16. [NAT · 1 mark]**

Find the single equivalent discount for successive discounts of 10% and 10%.

**Answer: 19**

**Solution:** Equivalent discount =  $10 + 10 - (10 \times 10)/100 = 20 - 1 = 19\%$ .

---

[New Practice]

**Q17. [MCQ · 2 marks]**

A shopkeeper marks his goods 25% above cost price and allows a discount of 20%. Find his overall profit or loss percentage.

- (A) 0% (no profit, no loss)
- (B) 5% profit
- (C) 5% loss
- (D) 10% profit

**Answer: (A)**

**Solution:** Let CP=100. MP=125. SP=125×0.8=100. Since SP=CP, there is no profit and no loss.

---

*[New Practice]*

**Q18. [NAT · 2 marks]**

A trader normally sells an item at an 8% profit. If instead he had bought it for 10% less (same selling price), what would his new profit percentage be?

**Answer: 20**

**Solution:** Let original CP = x. Original SP = 1.08x. New CP = 0.9x, same SP = 1.08x. New profit% =  $(1.08x - 0.9x) / 0.9x \times 100 = (0.18x / 0.9x) \times 100 = 20\%$ .

---

*[New Practice]*

**Q19. [MCQ · 2 marks]**

The marked price of an item is 50% above its cost price. What discount percentage should be offered to sell it at cost price (no profit, no loss)?

- (A) 33.33%
- (B) 50%
- (C) 25%
- (D) 40%

**Answer: (A)**

**Solution:** CP=100, MP=150. To sell at 100, required discount =  $(150 - 100) / 150 \times 100 = 33.33\%$ .

---

*[New Practice]*

**Q20. [NAT · 2 marks]**

A shopkeeper allows a discount of 10% on the marked price and still makes a profit of 8%. If the cost price is Rs. 500, find the marked price.

**Answer: 600**

**Solution:** Required SP =  $500 \times 1.08 = 540$ .  $MP \times 0.9 = 540 \Rightarrow MP = 540 / 0.9 = 600$ .

---

# Topic 5: Introduction to Complexity - Big-O / Theta / Omega

## Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CSE 2023, Question 44, concept]

### Q1. [MSQ · 2 marks]

Let  $f_1(n)$  and  $f_2(n)$  be functions such that  $f_1(n) \in \Theta(f_2(n))$ . Which of the following statements is/are necessarily TRUE?

- (A)  $f_1(n) = f_2(n)$  exactly, for all  $n$
- (B)  $f_1(n) \in O(f_2(n))$
- (C)  $f_1(n) \in \Omega(f_2(n))$
- (D)  $f_1(n) \in O(n)$ , given that  $f_2(n) = O(n)$

**Answer: (B), (C), (D)**

**Solution:** By definition,  $f_1 \in \Theta(f_2)$  means  $f_1 \in O(f_2)$  AND  $f_1 \in \Omega(f_2)$ , so (B) and (C) always hold. (D) follows by transitivity of  $O$ . (A) is false --  $\Theta$  only guarantees the same growth rate up to constant factors, not exact equality (e.g.  $f_1(n)=2n$  and  $f_2(n)=n$  satisfy  $f_1 \in \Theta(f_2)$  but  $f_1 \neq f_2$ ).

[VERIFIED PYQ — GATE CS 2014, Set 1, Question 20]

### Q2. [MCQ · 2 marks]

Let  $G$  be a graph with  $n$  vertices and  $m$  edges, represented using an adjacency matrix. What is the tightest upper bound on the running time of Depth-First Search (DFS) on  $G$ ?

- (A)  $O(n)$
- (B)  $O(m+n)$
- (C)  $O(n^2)$
- (D)  $O(mn)$

**Answer: (C)**

**Solution:** With an adjacency-matrix representation, examining all possible neighbours of each of the  $n$  vertices requires scanning its entire row ( $n$  entries), regardless of the actual number of edges, giving a tight bound of  $O(n^2)$  overall -- as opposed to  $O(m+n)$  with an adjacency-list representation.

[GATE-style]

### Q3. [MCQ · 1 mark]

$f(n) = O(g(n))$  formally means there exist positive constants  $c$  and  $n_0$  such that:

- (A)  $f(n) \leq c \cdot g(n)$  for all  $n \geq n_0$
- (B)  $f(n) \geq c \cdot g(n)$  for all  $n \geq n_0$
- (C)  $f(n) = c \cdot g(n)$  for all  $n$
- (D)  $f(n) < g(n)$  for all  $n$

**Answer: (A)**

[GATE-style]

### Q4. [MCQ · 1 mark]

$f(n) = \Omega(g(n))$  formally means there exist positive constants  $c$  and  $n_0$  such that:

- (A)  $f(n) \leq c \cdot g(n)$  for all  $n \geq n_0$
- (B)  $f(n) \geq c \cdot g(n)$  for all  $n \geq n_0$
- (C)  $f(n) = c \cdot g(n)$  for all  $n$

(D)  $f(n) > g(n)$  for all  $n$

**Answer: (B)**

---

[GATE-style]

**Q5. [MCQ · 1 mark]**

$f(n) = \Theta(g(n))$  means:

- (A)  $f(n) = O(g(n))$  only
- (B)  $f(n) = \Omega(g(n))$  only
- (C)  $f(n) = O(g(n))$  AND  $f(n) = \Omega(g(n))$
- (D)  $f(n) = O(g(n))$  OR  $f(n) = \Omega(g(n))$

**Answer: (C)**

---

[GATE-style]

**Q6. [NAT · 1 mark]**

For  $f(n) = 3n^2 + 5n + 2$ , the tightest bound is  $\Theta(n^k)$ . Find  $k$ .

**Answer: 2**

---

[GATE-style]

**Q7. [MCQ · 2 marks]**

If  $f(n) = O(n^2)$  and  $g(n) = O(n^3)$ , then  $f(n) + g(n)$  is:

- (A)  $O(n^2)$
- (B)  $O(n^3)$
- (C)  $O(n^5)$
- (D)  $O(n^6)$

**Answer: (B)**

**Solution:** The sum is bounded by the growth rate of the dominant (faster-growing) term, which is  $O(n^3)$ .

---

[GATE-style]

**Q8. [MSQ · 2 marks]**

Which of the following statements are TRUE?

- (A)  $n \log n = O(n^2)$
- (B)  $n^2 = O(n \log n)$
- (C)  $2^n = O(n^{100})$
- (D)  $n! = \Omega(2^n)$  for large  $n$

**Answer: (A), (D)**

**Solution:** (B) is false --  $n^2$  grows strictly faster than  $n \log n$ . (C) is false -- the exponential  $2^n$  eventually dominates any fixed polynomial  $n^{100}$ .

---

[GATE-style]

**Q9. [MCQ · 1 mark]**

Which asymptotic notation gives a tight bound on a function's growth rate, both from above and below?

- (A)  $O$
- (B)  $\Omega$
- (C)  $\Theta$

(D)  $o$

**Answer: (C)**

---

*[GATE-style]*

**Q10. [MCQ · 1 mark]**

Which of these growth rates is the SLOWEST as  $n \rightarrow \infty$ ?

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n \log n)$
- (D)  $O(n^2)$

**Answer: (B)**

---

## Section B — New Practice Questions (10)

*[New Practice]*

**Q11. [MCQ · 1 mark]**

For the constant function  $f(n) = 5$ , the tightest bound is:

- (A)  $\Theta(n)$
- (B)  $\Theta(1)$
- (C)  $\Theta(\log n)$
- (D)  $\Theta(n^2)$

**Answer: (B)**

---

*[New Practice]*

**Q12. [NAT · 1 mark]**

For  $f(n) = 7n^3 + 2n^2 + 100$ , find  $k$  such that  $f(n) = \Theta(n^k)$ .

**Answer: 3**

---

*[New Practice]*

**Q13. [MCQ · 1 mark]**

Which of the following correctly orders these functions from slowest to fastest growing?

- (A)  $n, \log n, n^2, 2^n$
- (B)  $\log n, n, n^2, 2^n$
- (C)  $2^n, n^2, n, \log n$
- (D)  $n^2, n, \log n, 2^n$

**Answer: (B)**

---

*[New Practice]*

**Q14. [MSQ · 2 marks]**

Which of the following statements about asymptotic notation are TRUE?

- (A) If  $f(n) = \Theta(g(n))$ , then  $g(n) = \Theta(f(n))$  (symmetry)
- (B) If  $f(n) = O(g(n))$  and  $g(n) = O(h(n))$ , then  $f(n) = O(h(n))$  (transitivity)
- (C)  $f(n) = O(g(n))$  implies  $g(n) = O(f(n))$  always
- (D)  $f(n) = o(g(n))$  implies  $f(n) = O(g(n))$

**Answer: (A), (B), (D)**

**Solution:** (C) is false -- Big-O is not symmetric (e.g.  $n = O(n^2)$ , but  $n^2$  is not  $O(n)$ ).

---

[New Practice]

**Q15. [MCQ · 1 mark]**

For an algorithm with worst-case running time  $T(n) = 100n + 50$ , the tightest asymptotic bound is:

- (A)  $\Theta(1)$
- (B)  $\Theta(n)$
- (C)  $\Theta(n^2)$
- (D)  $\Theta(n \log n)$

**Answer: (B)**

---

[New Practice]

**Q16. [NAT · 1 mark]**

Is the statement " $n^2 = O(n^3)$ " TRUE or FALSE? Answer with 1 for TRUE, 0 for FALSE.

**Answer: 1**

**Solution:**  $n^2$  grows no faster than  $n^3$  asymptotically, so the statement is true.

---

[New Practice]

**Q17. [MCQ · 2 marks]**

For the same pair of functions  $f(n)$  and  $g(n)$ , which relationship correctly connects Big-O and Big-Omega?

- (A)  $f(n) = O(g(n))$  if and only if  $g(n) = \Omega(f(n))$
- (B)  $f(n) = O(g(n))$  if and only if  $g(n) = O(f(n))$
- (C)  $f(n) = \Omega(g(n))$  if and only if  $f(n) = O(g(n))$
- (D) There is no relationship between  $O$  and  $\Omega$

**Answer: (A)**

**Solution:** By definition,  $f(n) = O(g(n))$  ( $f$  upper-bounded by  $g$  up to constants) is exactly equivalent to saying  $g$  is a lower bound for  $f$ , i.e.  $g(n) = \Omega(f(n))$ .

---

[New Practice]

**Q18. [MCQ · 1 mark]**

A loop running from  $i=1$  to  $n$ , doubling  $i$  each iteration ( $i = i \times 2$ ), has time complexity:

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n \log n)$
- (D)  $O(n^2)$

**Answer: (B)**

---

[New Practice]

**Q19. [NAT · 1 mark]**

Nested loops (an outer loop running  $n$  times, and for each outer iteration, an inner loop also running  $n$  times) have time complexity  $\Theta(n^k)$ . Find  $k$ .

**Answer: 2**

---

[New Practice]

**Q20. [MCQ · 2 marks]**

Which is the correct order of growth (from FASTEST to slowest) among  $O(n!)$ ,  $O(2^n)$ ,  $O(n^{10})$ ,  $O(\log n)$ ?

- (A)  $O(n!) > O(2^n) > O(n^{10}) > O(\log n)$
- (B)  $O(\log n) > O(n^{10}) > O(2^n) > O(n!)$
- (C)  $O(2^n) > O(n!) > O(n^{10}) > O(\log n)$
- (D)  $O(n^{10}) > O(n!) > O(2^n) > O(\log n)$

**Answer: (A)**

---

## Topic 6: Solving Recurrences via the Substitution Method

This topic includes four fully worked substitution-method problems, spread across the sections below: (1)  $T(n)=T(n-1)+n$  [linear, exact closed form], (2)  $T(n)=2T(n/2)+n$  [merge-sort-style,  $O(n \log n)$ ], (3)  $T(n)=T(n/2)+1$  [binary-search-style,  $\Theta(\log n)$ ], and (4)  $T(n)=3T(n/2)+n$  [Karatsuba-style,  $O(n^{\log_2 3})$ ]. Since none of these are GATE DA/CSE verbatim past questions, all are labelled [GATE-style], written in the authentic substitution/induction format GATE uses.

### Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

#### Q1. [MCQ · 1 mark]

The substitution method for solving a recurrence relation  $T(n)$  involves:

- (A) Directly computing  $T(n)$  for every  $n$  by brute force
- (B) Guessing a closed-form solution and then proving it correct by mathematical induction (substituting the guess back into the recurrence)
- (C) Always converting the recurrence into a recursion-tree diagram instead
- (D) Applying the Master theorem exclusively

**Answer: (B)**

---

[GATE-style, Substitution Problem 1:  $T(n)=T(n-1)+n$ ]

#### Q2. [MCQ · 2 marks]

Consider  $T(n) = T(n-1) + n$ , with  $T(1) = 1$ . Using the substitution/induction method with the guess  $T(n) = n(n+1)/2$ , and assuming (inductive hypothesis)  $T(n-1) = (n-1)n/2$ , which of the following correctly completes the inductive step?

- (A)  $T(n) = (n-1)n/2 + n = n(n+1)/2$ , matching the guess
- (B)  $T(n) = (n-1)n/2 + n = n^2/2$ , which does NOT match the guess
- (C)  $T(n) = (n-1)n/2 - n$
- (D)  $T(n)$  cannot be determined without further information

**Answer: (A)**

**Solution:**  $T(n) = T(n-1) + n = (n-1)n/2 + n = [n^2 - n + 2n]/2 = [n^2 + n]/2 = n(n+1)/2$ , exactly confirming the guessed closed form.

---

[GATE-style, Substitution Problem 1 continued]

#### Q3. [NAT · 1 mark]

Using  $T(n) = T(n-1) + n$ ,  $T(1) = 1$ , compute  $T(5)$  by direct unrolling.

**Answer: 15**

**Solution:**  $T(1)=1$ ,  $T(2)=3$ ,  $T(3)=6$ ,  $T(4)=10$ ,  $T(5)=15$ . (Matches the closed form  $n(n+1)/2 = 5 \times 6/2 = 15$ .)

---

[GATE-style, Substitution Problem 2:  $T(n)=2T(n/2)+n$ ]

#### Q4. [MCQ · 2 marks]

For  $T(n) = 2T(n/2) + n$ ,  $T(1)=1$ , the substitution method proves  $T(n) = O(n \log n)$  by guessing  $T(n) \leq cn \log n$  and substituting. Which inequality chain correctly completes the inductive step (for  $n \geq 2$ , assuming  $T(n/2) \leq c(n/2)\log(n/2)$ )?

- (A)  $T(n) = 2T(n/2) + n \leq 2c(n/2)\log(n/2) + n = cn \cdot \log(n/2) + n = cn \log n - cn + n \leq cn \log n$  (for  $c \geq 1$ )

- (B)  $T(n) = 2T(n/2) + n \leq cn \log n + n$ , which always exceeds  $cn \log n$   
(C)  $T(n) = 2T(n/2) + n \leq c(n/2)\log(n/2)$ , ignoring the  $+n$  term entirely  
(D)  $T(n)$  cannot be bounded by  $n \log n$  using substitution

**Answer: (A)**

---

[GATE-style, Substitution Problem 2 continued]

**Q5. [NAT · 2 marks]**

Using  $T(n) = 2T(n/2) + n$ ,  $T(1)=1$  ( $n$  a power of 2), compute  $T(8)$  by direct unrolling.

**Answer: 32**

**Solution:**  $T(1)=1$ .  $T(2)=2(1)+2=4$ .  $T(4)=2(4)+4=12$ .  $T(8)=2(12)+8=32$ .

---

[GATE-style, Substitution Problem 3:  $T(n)=T(n/2)+1$ ]

**Q6. [MCQ · 2 marks]**

For  $T(n) = T(n/2)+1$ ,  $T(1)=1$ , guessing  $T(n) = \log_2 n + 1$ , and assuming  $T(n/2) = \log_2(n/2)+1$ , which substitution correctly verifies the guess for  $n \geq 2$ ?

- (A)  $T(n) = T(n/2)+1 = [\log_2(n/2)+1]+1 = (\log_2 n - 1)+2 = \log_2 n + 1$ , matching the guess  
(B)  $T(n) = T(n/2)+1 = \log_2 n + 2$ , which does not match  
(C)  $T(n) = T(n/2)+1 = \log_2(n/2)$   
(D) The guess is incorrect and cannot be verified

**Answer: (A)**

**Solution:**  $\log_2(n/2) = \log_2 n - 1$ , so  $T(n) = (\log_2 n - 1 + 1) + 1 = \log_2 n + 1$ , exactly matching the guessed form.

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[GATE-style, Substitution Problem 3 continued]

**Q7. [NAT · 1 mark]**

Using  $T(n) = T(n/2)+1$ ,  $T(1)=1$  ( $n$  a power of 2), compute  $T(16)$  by direct unrolling.

**Answer: 5**

**Solution:**  $T(1)=1$ ,  $T(2)=2$ ,  $T(4)=3$ ,  $T(8)=4$ ,  $T(16)=5$ . (Matches the formula  $\log_2 16 + 1 = 4 + 1 = 5$ .)

---

[GATE-style, Substitution Problem 4:  $T(n)=3T(n/2)+n$ ]

**Q8. [MCQ · 2 marks]**

For  $T(n) = 3T(n/2) + n$ ,  $T(1)=1$ , the substitution method guesses  $T(n) = O(n^{\log_2 3})$ . What is the approximate numeric value of  $\log_2 3$  used as the exponent (correct to 2 decimals)?

- (A) 1.58  
(B) 1.50  
(C) 2.00  
(D) 3.00

**Answer: (A)**

**Solution:**  $\log_2 3 = \ln 3 / \ln 2 \approx 1.0986 / 0.6931 \approx 1.585$ .

---

[GATE-style, Substitution Problem 4 continued]

**Q9. [NAT · 2 marks]**

Using  $T(n) = 3T(n/2) + n$ ,  $T(1)=1$  ( $n$  a power of 2), compute  $T(4)$  by direct unrolling.

**Answer: 19**

**Solution:**  $T(1)=1$ .  $T(2)=3(1)+2=5$ .  $T(4)=3(5)+4=19$ .

---

[GATE-style]

**Q10. [MSQ · 2 marks]**

Which of the following are TRUE about the substitution method for solving recurrences?

- (A) It requires first guessing the form of the solution
- (B) The guess is verified using mathematical induction
- (C) It always gives an exact closed-form solution rather than an asymptotic bound
- (D) A common pitfall is trying to prove a bound that is asymptotically too weak or too strong without adjusting the constants appropriately

**Answer: (A), (B), (D)**

**Solution:** (C) is false -- the substitution method is often used specifically to establish asymptotic ( $O$ ,  $\Omega$ , or  $\Theta$ ) bounds rather than an exact formula, although (as in Problem 1) it can sometimes yield an exact closed form too.

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**Section B — New Practice Questions (10)**

*[New Practice]*

**Q11. [NAT · 1 mark]**

$T(n) = T(n-1) + 2$ ,  $T(1) = 3$ . Compute  $T(4)$  by direct unrolling.

**Answer: 9**

**Solution:**  $T(1)=3$ ,  $T(2)=5$ ,  $T(3)=7$ ,  $T(4)=9$ .

---

*[New Practice]*

**Q12. [MCQ · 2 marks]**

For  $T(n)=T(n-1)+2$ ,  $T(1)=3$ , the closed-form solution (verifiable by substitution) is:

- (A)  $T(n) = 2n+1$
- (B)  $T(n) = n+2$
- (C)  $T(n) = 2n$
- (D)  $T(n) = n^2+2$

**Answer: (A)**

**Solution:** Unrolling gives  $T(n) = 3+2(n-1) = 2n+1$ . Check:  $T(1)=3 \checkmark$ ,  $T(4)=9 \checkmark$ .

---

*[New Practice]*

**Q13. [NAT · 2 marks]**

$T(n) = 4T(n/2) + n$ ,  $T(1) = 1$ . Compute  $T(4)$  by direct unrolling.

**Answer: 28**

**Solution:**  $T(1)=1$ .  $T(2)=4(1)+2=6$ .  $T(4)=4(6)+4=28$ .

---

*[New Practice]*

**Q14. [MCQ · 2 marks]**

For  $T(n) = 4T(n/2) + n$ , the substitution method would be used to guess and verify a bound of order:

- (A)  $O(n)$
- (B)  $O(n \log n)$
- (C)  $O(n^2)$
- (D)  $O(2^n)$

**Answer: (C)**

**Solution:** Here  $a=4$ ,  $b=2$ , and  $n^{\log_2 a} = n^2$  dominates the additive  $n$  term, so the recurrence solves to  $\Theta(n^2)$ ; substitution would verify a guess of the form  $T(n) \leq cn^2 - dn$ .

---

[New Practice]

**Q15. [NAT · 2 marks]**

$T(n) = T(n-1) + n^2$ ,  $T(1) = 1$ . Compute  $T(4)$  by direct unrolling.

**Answer: 30**

**Solution:**  $T(1)=1$ .  $T(2)=1+4=5$ .  $T(3)=5+9=14$ .  $T(4)=14+16=30$ .

---

[New Practice]

**Q16. [MSQ · 2 marks]**

When applying the substitution method to prove  $T(n) \leq cf(n)$  by induction, which of the following are essential parts of a correct proof?

- (A) A base case verifying the bound holds for small  $n$
- (B) An inductive hypothesis assuming the bound holds for all values smaller than  $n$
- (C) An inductive step showing the bound then holds for  $n$  itself, using the recurrence
- (D) Skipping the base case is acceptable as long as the inductive step works

**Answer: (A), (B), (C)**

**Solution:** (D) is false -- omitting the base case is a serious and common error; without it, the induction is not actually established.

---

[New Practice]

**Q17. [NAT · 1 mark]**

$T(n) = 2T(n-1) + 1$ ,  $T(1) = 1$ . Compute  $T(4)$  by direct unrolling.

**Answer: 15**

**Solution:**  $T(1)=1$ .  $T(2)=2(1)+1=3$ .  $T(3)=2(3)+1=7$ .  $T(4)=2(7)+1=15$ .

---

[New Practice]

**Q18. [MCQ · 2 marks]**

The closed-form solution to  $T(n) = 2T(n-1)+1$ ,  $T(1)=1$  is:

- (A)  $T(n) = 2^n - 1$
- (B)  $T(n) = 2^n$
- (C)  $T(n) = n^2$
- (D)  $T(n) = 2n-1$

**Answer: (A)**

**Solution:** This is the classic Tower-of-Hanoi-style recurrence,  $T(n) = 2^n - 1$ . Check:  $T(4) = 16 - 1 = 15$  ✓, matching the previous question.

---

[New Practice]

**Q19. [NAT · 1 mark]**

$T(n) = T(n/3) + 1$ ,  $T(1) = 1$  ( $n$  a power of 3). Compute  $T(9)$  by direct unrolling.

**Answer: 3**

**Solution:**  $T(1)=1$ .  $T(3)=T(1)+1=2$ .  $T(9)=T(3)+1=3$ .

---

[New Practice]

**Q20. [MCQ · 2 marks]**

For  $T(n)=T(n/3)+1$ ,  $T(1)=1$ , the closed form (verified via substitution) is:

- (A)  $T(n) = \log_3 n + 1$

(B)  $T(n) = 3 \log n$

(C)  $T(n) = n/3$

(D)  $T(n) = \log_2 n + 1$

**Answer: (A)**

**Solution:** Check with  $n=9$ :  $\log_3 9 + 1 = 2 + 1 = 3$ , matching the computed  $T(9)=3$  from the previous question.

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# Topic 7: Uninformed Search - DFS: Mechanism & Properties

## Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CSE 2025, Set 2, Question 19]

### Q1. [MSQ · 2 marks]

Which of the following statements regarding Breadth-First Search (BFS) and Depth-First Search (DFS) on an undirected simple graph  $G$  is/are TRUE? (i) A DFS tree of  $G$  is a shortest-path tree of  $G$ . (ii) Every non-tree edge of  $G$  with respect to a DFS tree is a forward/back edge. (iii) If  $(u,v)$  is a non-tree edge of  $G$  with respect to a BFS tree, then the distances from the source to  $u$  and to  $v$  in the BFS tree are within 1 of each other. (iv) Both BFS and DFS can be used to find the connected components of  $G$ .

- (A) (i) only
- (B) (ii), (iii) and (iv)
- (C) (i), (ii) and (iii)
- (D) All four statements

### Answer: (B)

**Solution:** (i) is FALSE in general -- a DFS tree does not guarantee shortest paths (DFS can reach a vertex via a longer route even when a shorter one exists), unlike a BFS tree. (ii) is TRUE -- in an undirected graph, DFS never produces cross edges; every non-tree edge connects an ancestor-descendant pair (a back edge, viewable as forward/back from either endpoint). (iii) is TRUE -- a standard property of BFS trees on undirected graphs. (iv) is TRUE -- repeating either traversal from every unvisited vertex correctly finds all connected components.

[GATE-style]

### Q2. [MCQ · 1 mark]

DFS explores a graph by:

- (A) Visiting all neighbours of a node before moving to the next level (level-by-level)
- (B) Going as deep as possible along one branch before backtracking
- (C) Always choosing the node with the lowest heuristic value
- (D) Visiting nodes in a purely random order

### Answer: (B)

[GATE-style]

### Q3. [MCQ · 1 mark]

DFS is naturally implemented using which mechanism (either explicitly, or implicitly via the language's function-call mechanism)?

- (A) A queue (FIFO)
- (B) A stack (LIFO), or equivalently recursion (which itself uses the call stack)
- (C) A priority queue
- (D) A circular buffer

### Answer: (B)

[GATE-style]

### Q4. [MCQ · 1 mark]

The time complexity of DFS on a graph with  $V$  vertices and  $E$  edges, represented using an adjacency list, is:

- (A)  $O(V)$
- (B)  $O(V+E)$
- (C)  $O(V \times E)$
- (D)  $O(V^2)$ , always, regardless of representation

**Answer: (B)**

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[GATE-style]

**Q5. [MCQ · 1 mark]**

The space complexity of DFS (for the recursion / explicit stack, excluding the graph's own storage), in the worst case, is:

- (A)  $O(1)$
- (B)  $O(V)$  -- bounded by the maximum recursion depth, up to the number of vertices
- (C)  $O(E)$
- (D)  $O(V \times E)$

**Answer: (B)**

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[GATE-style]

**Q6. [MSQ · 2 marks]**

Which of the following properties hold for DFS on a finite, connected graph?

- (A) DFS is complete (guaranteed to visit every reachable vertex) on a finite graph
- (B) DFS is generally NOT guaranteed to find the shortest path (fewest edges) to a goal, unlike BFS
- (C) DFS typically uses less memory than BFS in the worst case, since it need not store an entire frontier level
- (D) DFS can get stuck in an infinite loop on an infinite (or cyclic, if visited-tracking is omitted) search space

**Answer: (A), (B), (C), (D)**

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[GATE-style]

**Q7. [MCQ · 1 mark]**

In DFS, a "back edge" connects the current vertex to:

- (A) An unvisited vertex
- (B) An ancestor of the current vertex in the DFS tree
- (C) A vertex in a completely different, already fully-explored subtree
- (D) The root of the DFS tree only

**Answer: (B)**

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[GATE-style]

**Q8. [MCQ · 2 marks]**

Iterative Deepening DFS (repeatedly running depth-limited DFS with increasing depth limits) is used primarily to:

- (A) Reduce the time complexity of DFS below  $O(V+E)$
- (B) Combine DFS's low memory usage ( $O(d)$ , where  $d$  is the current depth limit) with BFS's completeness and optimality guarantees (for equal step costs)
- (C) Eliminate the need for a visited-set entirely

(D) Guarantee finding the maximum-cost path

**Answer: (B)**

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*[GATE-style]*

**Q9. [MCQ · 1 mark]**

A key disadvantage of plain (depth-unlimited) DFS when searching a space with an infinite or very deep branch containing no solution is that it:

- (A) Uses too much memory, similar to BFS
- (B) Can get "stuck" exploring that infinite/deep branch and may never find a shallower goal that exists elsewhere, unless depth-limited
- (C) Is guaranteed to remain optimal regardless of depth
- (D) Cannot be implemented recursively

**Answer: (B)**

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*[GATE-style]*

**Q10. [MCQ · 1 mark]**

Compared to BFS, DFS is generally preferred when:

- (A) The shortest path (fewest edges) to the goal must be guaranteed
- (B) Memory is severely constrained and any valid solution (not necessarily the shortest) is acceptable
- (C) All step costs must be treated as strictly equal
- (D) The branching factor is infinite and depth is unbounded, with no depth limit imposed at all

**Answer: (B)**

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## Section B — New Practice Questions (10)

*[New Practice]*

**Q11. [MCQ · 1 mark]**

Which auxiliary data structure prevents DFS from repeatedly revisiting the same node on a graph with cycles?

- (A) A stack of visited nodes that is never actually checked
- (B) A visited/explored set (hash set or boolean array)
- (C) No such structure is needed
- (D) A priority queue

**Answer: (B)**

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*[New Practice]*

**Q12. [NAT · 2 marks]**

DFS is run on a tree with branching factor  $b=3$  and maximum depth  $d=4$  (edges from root to deepest leaf). What is the maximum number of nodes on the DFS recursion stack at the deepest point (counting the root)?

**Answer: 5**

**Solution:** A node at depth 4 has ancestors at depths 0,1,2,3, plus itself at depth 4 -- 5 nodes total on the stack.

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*[New Practice]*

**Q13. [MCQ · 1 mark]**

DFS applied to a Directed Acyclic Graph (DAG) can be used to compute:

- (A) The shortest path between all pairs of vertices directly
- (B) A topological ordering of the vertices
- (C) The minimum spanning tree
- (D) The maximum flow

**Answer: (B)**

---

*[New Practice]*

**Q14. [MSQ · 2 marks]**

Which of the following are TRUE about DFS edge classification in a directed graph?

- (A) Tree edges are edges that are part of the DFS forest/tree itself
- (B) Back edges connect a vertex to an ancestor in the DFS tree, and their presence indicates a cycle
- (C) Forward edges connect a vertex to a proper (non-child) descendant in the DFS tree
- (D) Cross edges never occur when running DFS on an undirected graph

**Answer: (A), (B), (C), (D)**

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*[New Practice]*

**Q15. [MCQ · 1 mark]**

If DFS is run on a directed graph and no back edge is ever found, what can be concluded about the graph?

- (A) The graph is a tree
- (B) The graph is a Directed Acyclic Graph (DAG), i.e. it contains no cycles
- (C) The graph is disconnected
- (D) The graph has exactly one vertex

**Answer: (B)**

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*[New Practice]*

**Q16. [NAT · 1 mark]**

A DFS traversal starting from vertex S visits all 6 vertices of a connected graph exactly once (backtracking as needed). How many vertices are in the resulting DFS tree?

**Answer: 6**

**Solution:** A full DFS traversal of a connected graph includes every vertex exactly once in its DFS tree.

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*[New Practice]*

**Q17. [MCQ · 2 marks]**

Which statement correctly compares DFS's completeness with BFS's completeness, on a finite graph?

- (A) Both DFS and BFS are complete (will find a solution if one exists) on any finite graph
- (B) Only BFS is complete; DFS is never complete
- (C) Only DFS is complete; BFS is never complete
- (D) Neither is complete on any graph

**Answer: (A)**

**Solution:** On a finite graph, both traversals eventually explore every reachable vertex, so both are complete; the important distinctions between them concern optimality and memory usage, not completeness.

---

[New Practice]

**Q18. [MCQ · 1 mark]**

DFS-based algorithms are commonly used for which of the following classic graph problems?

- (A) Finding strongly connected components (e.g. via Kosaraju's or Tarjan's algorithm)
- (B) Detecting cycles in a graph
- (C) Solving mazes/puzzles that have a single path to a solution
- (D) All of the above

**Answer: (D)**

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[New Practice]

**Q19. [NAT · 2 marks]**

DFS visits the nodes of a complete binary tree of depth 3 (i.e. depths 0,1,2,3 have 1,2,4,8 nodes respectively, 15 nodes total). What is the maximum size of the DFS call stack (counting the root as depth 0, so stack size = max depth + 1)?

**Answer: 4**

**Solution:** Maximum depth = 3, so the maximum stack size =  $3+1 = 4$ .

---

[New Practice]

**Q20. [MCQ · 2 marks]**

A key trade-off between DFS and BFS is that:

- (A) DFS typically uses less memory ( $O(\text{depth})$ ) but is not guaranteed to find the shortest path, while BFS uses more memory ( $O(\text{branching factor}^{\text{depth}})$ ) but guarantees the shortest path in unweighted graphs
- (B) DFS always uses more memory than BFS
- (C) BFS never guarantees finding a solution, while DFS always does
- (D) There is no meaningful trade-off between the two

**Answer: (A)**

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